Real Estate Price Prediction Report

1. Project Overview

Goal: Predict house market value (Price) using a combination of traditional tabular features (size, rooms, quality, location) and satellite imagery. This report documents the complete end-to-end pipeline implemented in the notebook.

- Inputs (Train): tabular features + target price
- Inputs (Test): same tabular features (no price) → generate predicted_price
- Models built: (A) Tabular baseline and (B) Multimodal CNN + tabular fusion (built but worse metrics)

2. Data Loading & Target Transformation

The dataset contains property-level records with numeric features (beds/baths, living area, lot size, quality grade, and location coordinates). The target variable is price.

Because price is heavily right-skewed, the notebook trains models on log1p(price) to stabilize variance and reduce the influence of very expensive outliers. Final predictions are converted back using expm1.

3. Exploratory Data Analysis

EDA focuses on: (i) understanding the price distribution, (ii) identifying which features most affect price, and (iii) checking relationships among features (multicollinearity) and location patterns.

3.1 Target distribution (raw vs log1p)

Raw price is strongly right-skewed (long tail). After log1p transform, the distribution becomes more bell-shaped and easier to model with common regression algorithms.

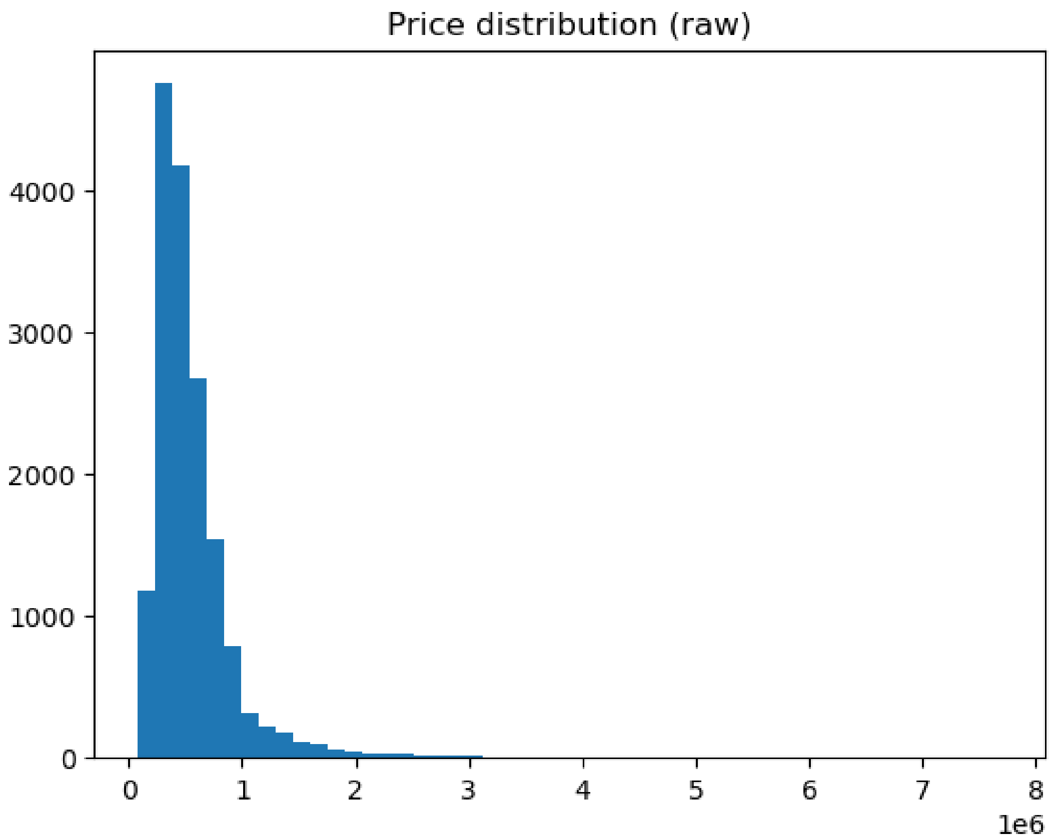

Figure: Price distribution (raw) — right-skewed.

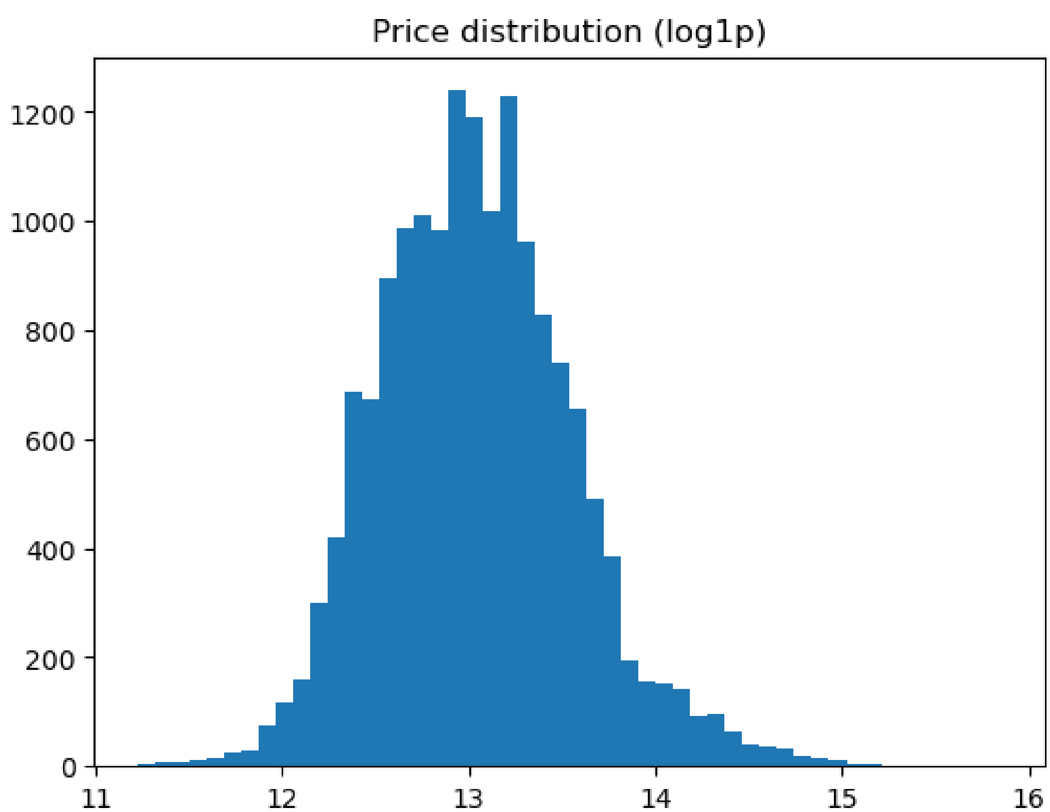

Figure: Price distribution (log1p) — closer to normal.

3.2 Feature impact on Price

Correlation with price (top drivers):

Feature	Corr(price)
sqft_living	0.701
grade	0.664
sqft_above	0.603
sqft_living15	0.582
bathrooms	0.525
view	0.391
sqft_basement	0.32
lat	0.31
bedrooms	0.304
floors	0.251

Interpretation (from the correlations and scatter plots in the notebook): larger living area and higher construction grade are the strongest linear indicators of higher prices.

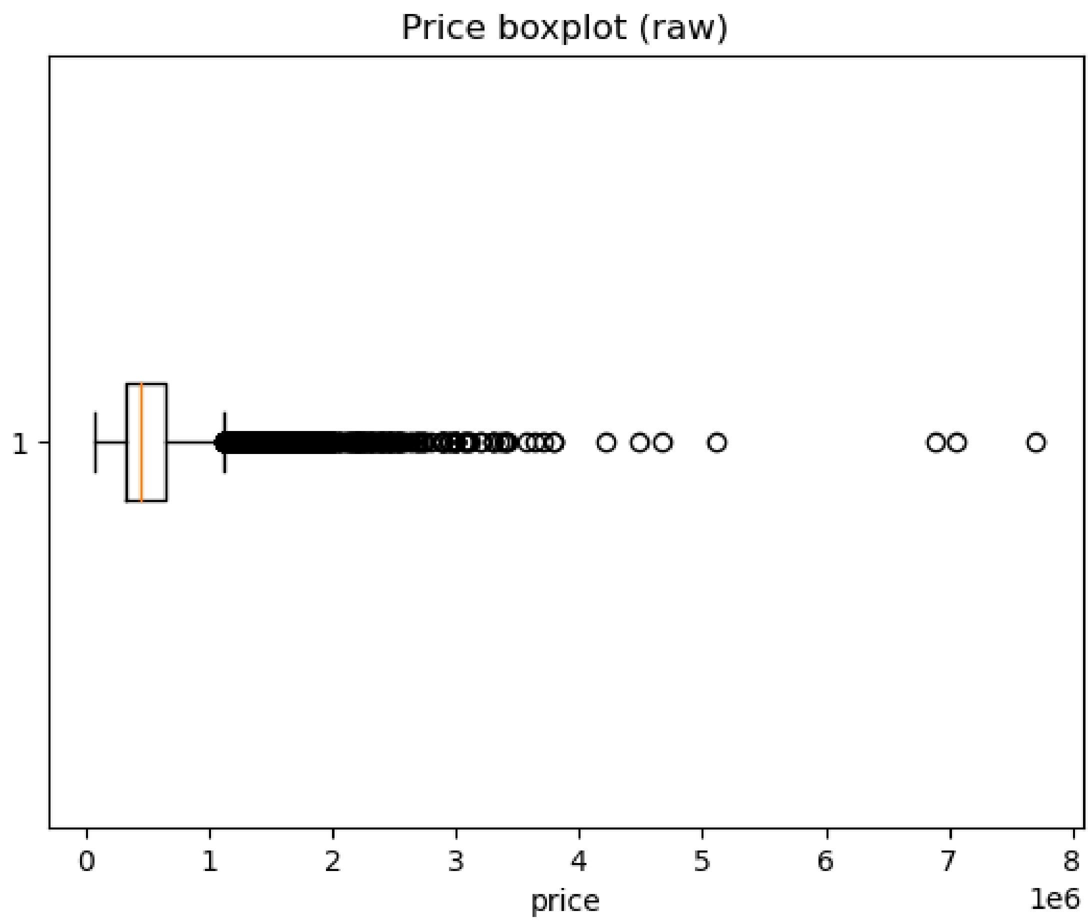

Figure: Correlation of key features with Price (notebook plot).

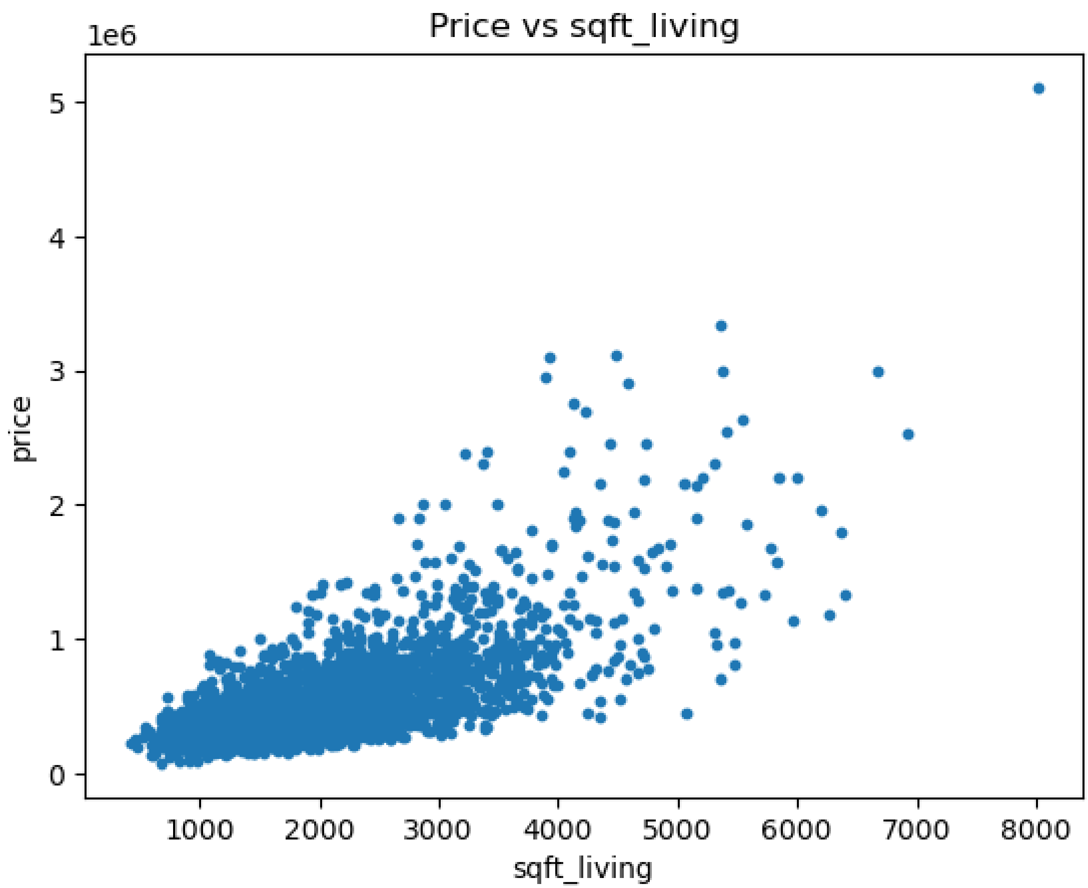

Figure: Example scatter check — Price vs sqft_living.

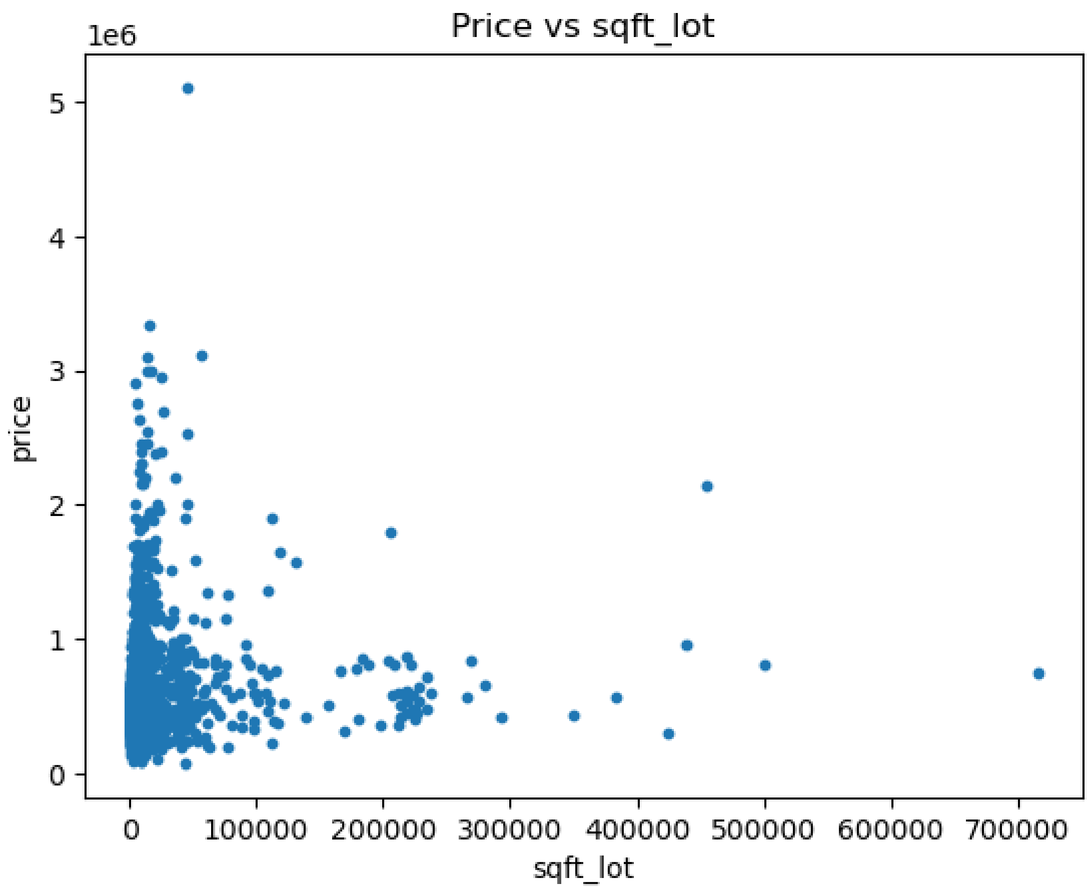

Figure: Example scatter check — Price vs grade (notebook plot).

3.3 Relationships among features

Several predictors are strongly correlated with each other (e.g., sqft_living vs sqft_above). This indicates multicollinearity — tree-based models handle this well, but linear models may require care.

Feature A	Feature B	Correlation
sqft_living	sqft_above	0.876
sqft_living	grade	0.761
sqft_living	sqft_living15	0.755
grade	sqft_above	0.753
bathrooms	sqft_living	0.752
sqft_above	sqft_living15	0.73
sqft_lot	sqft_lot15	0.721
grade	sqft_living15	0.707

3.4 Location and zipcode effects

Location affects price: properties cluster into high-price and low-price regions. Zipcode also acts as a strong neighborhood proxy (captures school quality, amenities, and demand).

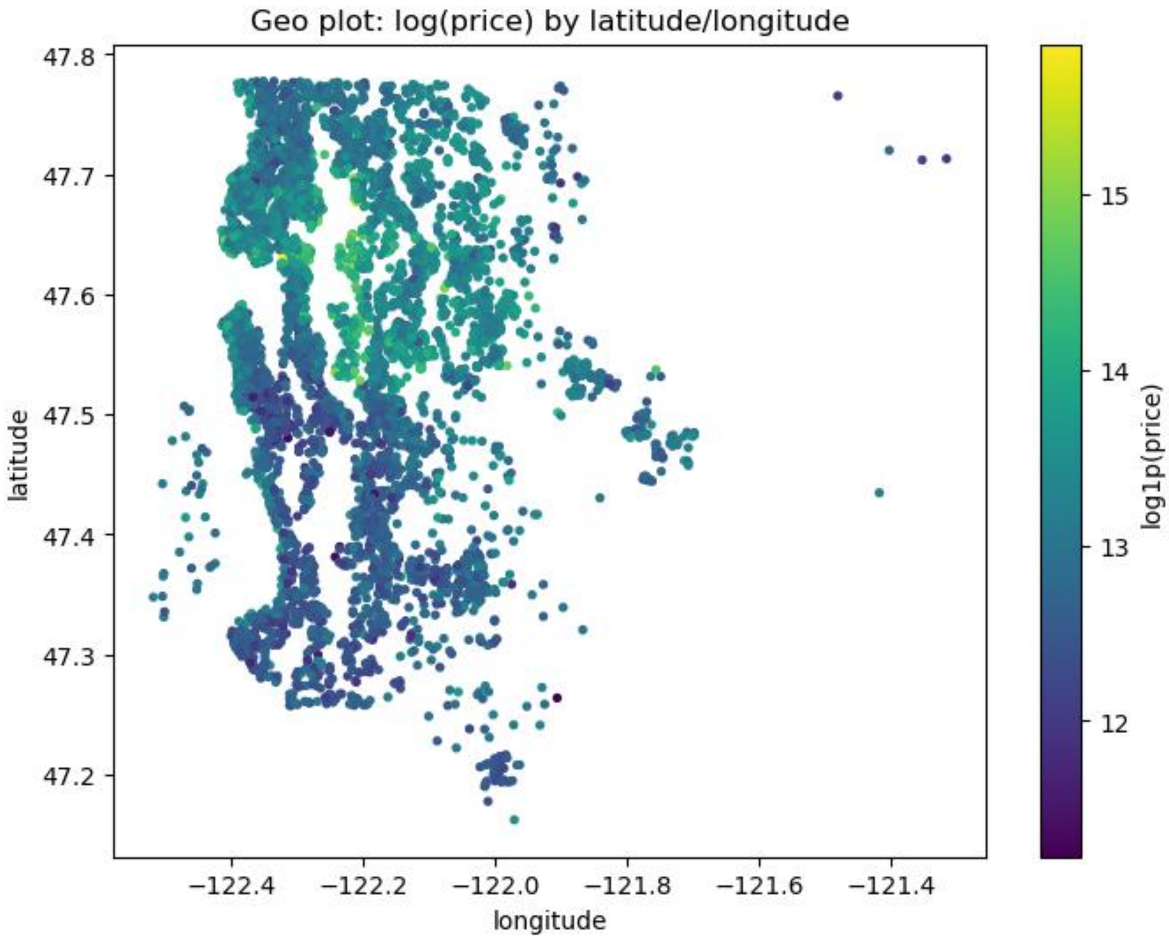

Figure: Spatial scatter of prices using latitude/longitude (notebook plot).

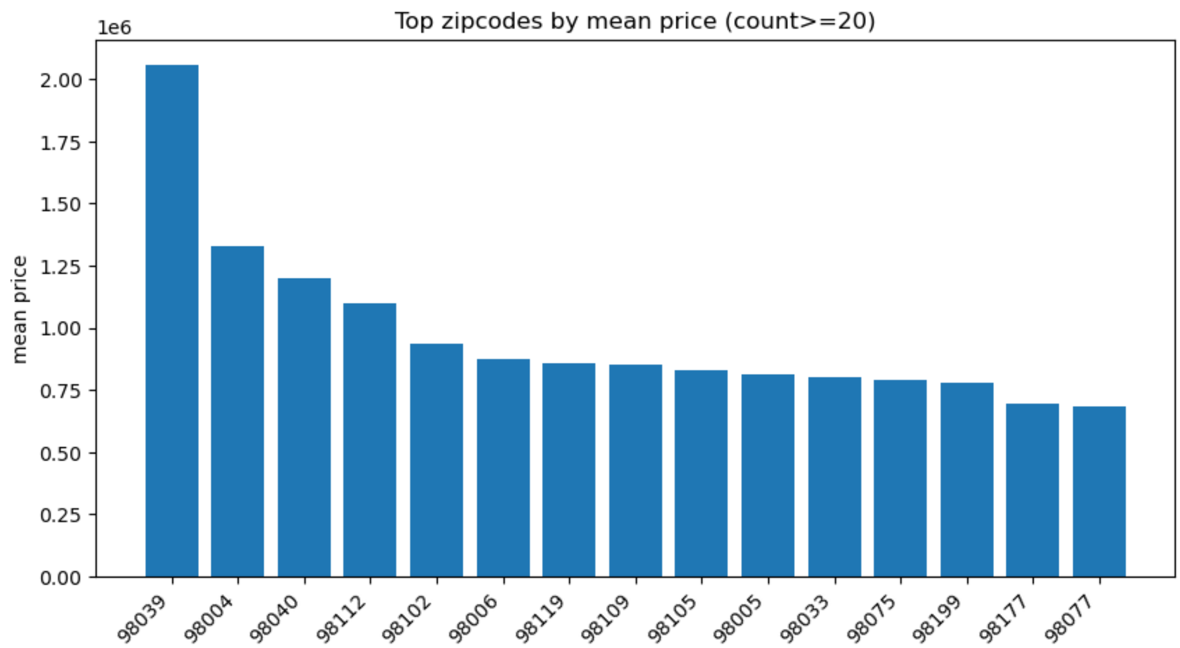

Figure: Top zipcodes by mean price (notebook plot).

3.5 Outliers (EDA check)

Boxplots and distribution checks showed strong outliers in area-related features (living area, lot size, etc.). These can dominate the loss and harm generalization if left untreated.

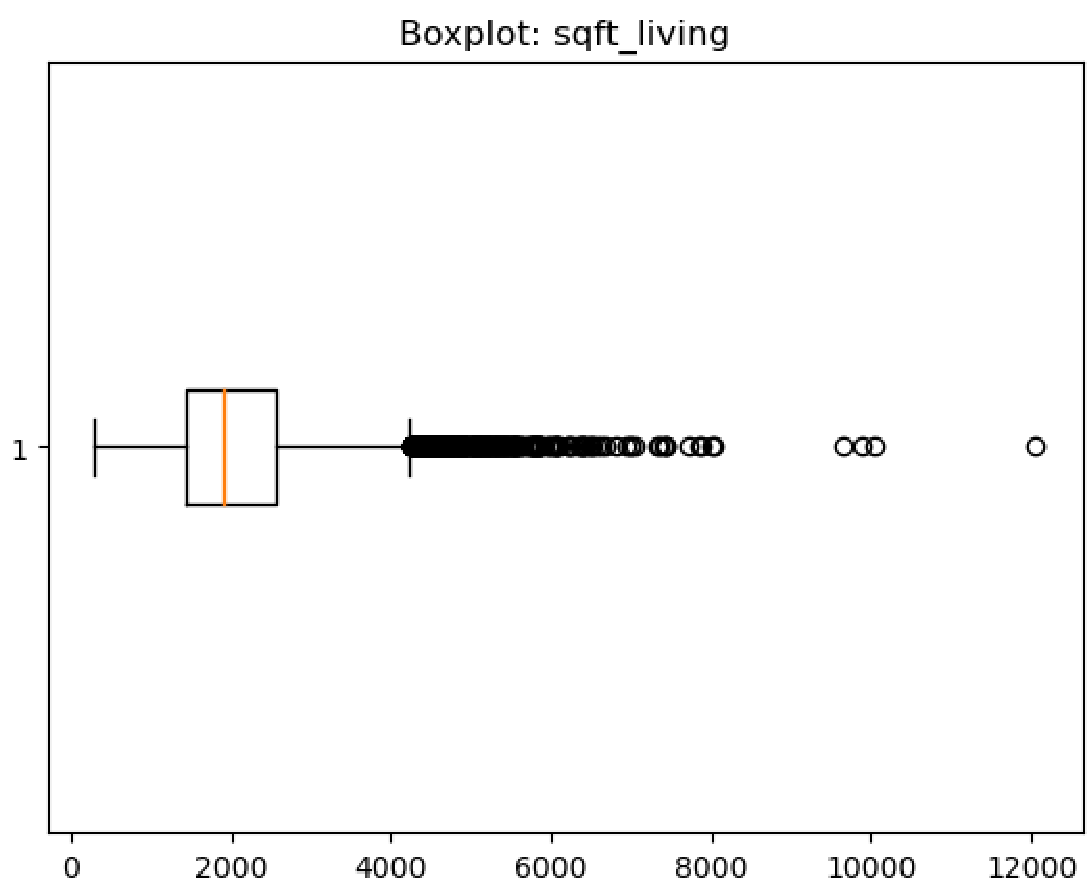

Figure: Boxplot example showing outliers in sqft_living.

4. Feature Engineering

The notebook creates additional features to capture meaningful ratios and interactions:

- above_to_living = sqft_above / sqft_living
- basement_to_living = sqft_basement / sqft_living
- bed_bath_interaction = bedrooms × bathrooms
- house_age = year(date) − yr_built
- is_renovated = 1 if yr_renovated > 0 else 0

These features help the model learn: layout composition (above vs basement), intensity of rooms, and age/renovation effects which are often non-linear.

5. Outlier Handling (Winsorisation / Quantile Clipping)

Instead of removing rows, the notebook caps extreme values for the strongest area-related features using 1st and 99th percentiles (computed from the training set). This keeps all samples while reducing the influence of extreme outliers.

- sqft_living: 1%≈720, 99%≈4920.30
- sqft_lot: 1%≈1005, 99%≈213009.95
- sqft_above: 1%≈700, 99%≈4380
- sqft_basement: 1%≈0, 99%≈1650
- sqft_living15: 1%≈950, 99%≈4042.24
- sqft_lot15: 1%≈1188, 99%≈169559.05

6. Encoding, Missing Values & Final Feature Matrix

- zipcode converted to string and One-Hot Encoded (OHE).
- OHE configured with handle_unknown='ignore' so unseen test zipcodes do not crash inference.
- After feature engineering and OHE, any remaining NaNs are filled with median values.
- Final tabular feature vector size in the notebook: 99 features (confirmed by DataLoader shape).

7. Modeling & Validation Results

7.1 Tabular baseline: HistGradientBoostingRegressor

A strong baseline was trained using HistGradientBoostingRegressor on the tabular feature matrix with target log1p(price). This model performed best and was chosen for final submission.

7.2 Multimodal model: CNN + Tabular Fusion (built but worse)

A multimodal pipeline was implemented: satellite images were downloaded via Mapbox using (lat, long), a CNN extracted image embeddings, and these were fused with tabular features for regression. However, the notebook validation metrics were significantly worse than the tabular baseline, so the final submission uses the tabular model.

Validation metrics summary (from notebook validation split):

Model	Target	Validation RMSE (log)	Validation R^2 (log)	Decision

Tabular baseline (HistGradientBoostingRegressor)	log1p(price)	0.1633	0.9033	Used for final submission
Multimodal (CNN + Tabular Fusion)	log1p(price)	0.5464	-0.0819	Built, but worse metrics

Note: Negative R^2 for the multimodal model indicates it performed worse than predicting the mean of the target on the validation split. This typically happens when the CNN is under-trained, images are noisy/misaligned, or tabular signal dominates.

8. Final Training & Submission File

- Fit the selected tabular model on the full training data (all rows).
- Predict log1p(price) for the test dataset.
- Convert predictions back to price using expm1.
- Create CSV with columns: id, predicted_price.

9. Conclusion

The project successfully built a strong tabular price prediction model with high validation performance. A multimodal satellite-imagery pipeline was also implemented, but it underperformed on the current validation split. Therefore, the tabular baseline is used for submission, while the multimodal approach remains a future improvement area.